

PGT13 Mini Report

Maike Ni - Identify parts of code which need to be refactored

Pavat Poonpinij - Select refactoring techniques to be applied

Jesse Ssempijja - Apply refactoring techniques to the code

What refactoring techniques did you apply?

- Extract function
- Move function
- Parametrise function
- Decompose conditional

What was the reason for applying these techniques?

- Reduce the size of code chunk to improve readability
- Some functions should be in the same class
- Some functions are similar that they can be the same function
- Simplify the conditional logic to improve readability

How do you know that observable behaviour has not changed after refactoring?

- By running the test coverage after refactoring

What are the (potential) benefits and drawbacks of applying these techniques?

- Benefits: readability, easier to understand, helpful for finding bugs
- Drawbacks: the process can be time consuming if the program is large, and may introduce unexpected behaviours, so test coverage is required to make sure the program still works as it is supposed to

Overall, how do you feel about the refactoring process? Worth it? Why (or why not)?

- We think refactoring is important, as the process makes the code easier to understand and modify, and enables developers to find bugs more easily. Though it takes some time to apply the techniques and to ensure that the systems still work correctly, the refactored code is worth the effort.